

YEASIN MORSHED JOY

Software Engineer — Machine Learning

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Professional Summary

Final-year BSc student in Computer Science & Engineering specializing in machine learning and full-stack software development. Experienced in designing and evaluating ML classification models (XGBoost, CatBoost, LightGBM) and applying explainable AI techniques (SHAP, LIME), alongside building production-ready web applications using React, Node.js, Express, Django, and relational databases. Strong technical documentation, problem-solving, and project delivery skills.

Technical Skills

- **Programming:** Python, JavaScript, C, C++, Java
- **Web Development:** React, Node.js, Express.js, Django, Laravel, REST APIs, HTML5, CSS3, Bootstrap
- **Machine Learning:** Scikit-learn, XGBoost, CatBoost, LightGBM, TensorFlow, PyTorch, SHAP, LIME
- **Databases:** MySQL, SQLite
- **Tools & Platforms:** Git, GitHub, Docker, VS Code, n8n, Postman
- **Productivity:** Microsoft Office, Google Workspace, LaTeX (Overleaf)

Internship Experience

Frontend & Backend Developer Intern — GAOTek Inc.

Aug 2026 – Present

- Contribute to frontend and backend development tasks across company web applications as assigned by the engineering team.
- Work across the full stack, implementing features, fixing bugs, and supporting ongoing production releases.
- Collaborate with cross-functional development teams adhering to standardized code reviews and Git workflows.

Project Experience

Explainable AI-Based Home Loan Approval Prediction System — Final Year Design Project

- Built a machine-learning system to predict home loan approval from demographic, financial, employment, and loan-related data across 78,000 records.
- Trained and compared XGBoost, CatBoost, LightGBM, and Random Forest classifiers, achieving over 90% accuracy.
- Applied SHAP and LIME to generate per-prediction explanations, improving model interpretability for non-technical stakeholders.
- Performed data preprocessing, feature engineering, feature selection, and hyperparameter optimization.
- Incorporated fairness, bias evaluation, transparency, privacy, and data-security considerations into system design.

Employee Management System — Full-Stack Web Application

- Built a responsive full-stack CRUD platform to manage employee records with search, filtering, and role handling.
- Engineered an Express.js REST API with comprehensive server-side input validation and JWT-based authentication for secure admin actions.
- Integrated persistent relational database storage with automated schema seeding and realistic sample records.

Repository: [View Repository](#)

Local Store E-Commerce Platform — Full-Stack Web Application

- Developed a full-featured online store with product browsing, product details view, and a real-time responsive UI using React.
- Built session-persistent shopping cart operations and a secure administrative dashboard for adding, editing, and deleting inventory.
- Designed RESTful backend endpoints with structured input verification, error boundaries, and instant user notifications.

Repository: [View Repository](#)

Road Condition and Surface Type Classification

- Built an ML classification pipeline to identify road conditions and surface types across 144,000 road-image/sensor records.
- Applied ensemble-learning methods (XGBoost, CatBoost, LightGBM) for feature handling, training, and evaluation, reaching 88.89% accuracy.

Repository: [View Repository](#)

Civic Desk – Issue Reporting & Tracking System — Web Application

- Built a civic issue-reporting platform (Django) for potholes, broken streetlights, and garbage-related complaints.
- Designed complaint-management dashboards and admin-coordination tooling to track issue resolution end-to-end.
- Led feature planning, testing, and iterative development across 3 release cycles.

Repository: [View Repository](#)

Puurfect Pals – Pet Adoption Platform — *Academic Project, Django Web App*

- Built a pet-adoption web platform supporting listings, veterinary doctor information, and image uploads.
- Implemented role-based access control and structured user flows for adopters and shelter managers.
- Documented the repository and API structure for long-term maintainability.

Repository: [View Repository](#)

Smart Walk Assistance Robot — *Microprocessors & Embedded Systems Project*

- Built an assistance robot prototype to assist visually impaired individuals in detecting nearby obstacles.
- Integrated Arduino Uno R3, HC-SR04 ultrasonic sensors, an L293D motor driver, vibration feedback, and buzzer alert mechanisms.

Repository: [View Repository](#)

Education

BSc in Computer Science & Engineering — *University of Asia Pacific, Dhaka* — Jan 2023 – Present

Higher Secondary Certificate (HSC), GPA: 4.67 / 5 — *Lakshmipur Govt. College* — 2019 – 2021

Secondary School Certificate (SSC), GPA: 4.44 / 5 — *Principal Kazi Faruki School and College* — 2017 – 2019

Communication & Leadership

- Public speaking and technical presentation delivery
- Team collaboration across academic and agile project settings
- Active listening, structured idea articulation, and project coordination

Extracurricular Activities

- Lakshmipur Govt. College Debate Club – Member (2019–2022)
- ICPC Dhaka Regional Preliminary Contest – Participant (2023)
- PKFSC Sports Club – Member (2016–2018)

Languages

Bangla (Native), English (Proficient)